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Procedia Economics and Finance 23 (2015) 1451 – 1456

Procedia
Economics and Finance

www.elsevier.com/locate/procedia

2nd GLOBAL CONFERENCE on BUSINESS, ECONOMICS, MANAGEMENT and TOURISM, 30-31 October 2014, Prague, Czech Republic

The impact of M&A on company performance: Evidence from Croatia

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Abstract

In the recent years the importance and value of M&As have reached enormous proportions. However, while some studies reported beneficial influence of M&A on companies' success, the others found this influence to be neutral or even negative. Furthermore, the majority of studies analysed M&A activity in USA and UK, therefore this paper aims to detect the consequences of M&A on the performance of companies operating within under-researched European area i.e. Croatia. The analysis was conducted on Croatian companies that were involved in M&A activity in period from 2008 to 2011. The obtained results clearly indicated statistically insignificant differences in target companies' performances before and after M&A activity. The same results, i.e. statistically insignificant differences, were obtained even when performance of target companies (after acquisition) was compared to the peer companies. Finally, the results showed that despite the growing influence of foreign investors in Croatian economy, the majority of M&A activity was carried out by Croatian companies.

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Keywords: M&A, performance, target companies, peer companies, Croatia

1. Introduction

Globalization, financial crisis, economic trends as well as numerous other factors such as the pressure of investors and competitors impose the imperative of flexibility on companies in order to survive on the market. Hence the internal growth of companies is often considered too slow and not efficient enough. On the other side, the importance of external growth, especially in form of takeovers, significantly rises over the last decades. The

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Selection and/ peer-review under responsibility of Academic World Research and Education Center
doi:10.1016/S2212-5671(15)00351-2

recession left a significant trace on M&A deals so the number and the value of M&A transactions is lower than in 2007 when they reached a peak with value of announced M&As of approximately 5 000 bil. USD (Thomson Financial, IMAA analysis, 2014). However, although the number of M&A transactions nowadays declined when compared to year 2007, a value of M&A transactions on the worldwide level for the year 2013 improved when compared to values for 2009, when first consequences of crisis began to feel.

With the US remaining the most active country and important growing markets (such as markets in Latin America, China and India), Europe is still the second most active region regarding M&A activity. Although M&A transactions involve companies from all over the world, empirical researches have mostly been focused on US and Great Britain. Studies on M&A transactions involving Eastern European companies, especially Croatia as a new EU member are relatively scarce. Hence, this research paper strives to add to the body of knowledge on M&A transactions by offering a somewhat different approach to analysis of impact of M&A on companies and by exploring growing Croatian M&A market. Croatian companies belong to the market that becomes more interesting for investors since Croatia is now an EU member and its geographical position provides numerous possibilities regarding economic cooperation with neighbouring countries that are also expected to contribute to worldwide M&A activity in the future. Further, a favourable geographical position characterised by road, rail, water and air transport network as well as rich water and wood resources offer numerous investment opportunities. Therefore, it is justified to expect an increase in the number and value of takeovers involving Croatian companies.

The research analyses the impact of takeovers solely on the performance of target companies. The focus is set on target companies which continue to operate independently after the takeover; hence this paper belongs to a small niche of M&A empirical studies following the respective research path. Target companies' performance is measured by different indicators of profitability and along with the use of both absolute and relative indicators in order to draw an objective conclusion; a distinctive feature of this research is the use of a control group formed by following a multiple criteria. Although this paper provides an insight into performance of relatively rarely analysed Croatian target companies, hopefully it will add to better understanding of the influence of takeovers on target companies regardless of the used sample since it uses methodology applicable outside Croatian borders.

The paper proceeds as follows. Section 2 provides a literature review while data description, estimation procedure and main results are presented in Section 3. The conclusion is present in Section 4.

2. Literature review

For decades M&A activity has been attracting scientific curiosity and due to complexity of this form of investments different theoretical approaches can be taken while exploring takeovers. A comprehensive overview of research standpoints in literature on M&A activity is provided by Larsson and Finkelstein (1999) who distinguish between five research standpoints: a) strategic management, b) economics, c) finance, d) company's organization and e) human resources while exploring takeovers. Along with different initial research standpoints an empirical analysis on M&A can be performed on a company, industry or aggregate (country) level. With all the above stated in mind and the fact that studies often vary in type of analysed data and selected time span a conflicting results on the same research issue do not come as a surprise.

Since this research is directed towards analysis of influence of takeovers on profitability it is important to set a suitable framework for an overview of the relevant literature. Bruner (2004) distinguishes between four research approaches:

- Event studies. They examine the abnormal returns to shareholders and therefore directly measure a value created for investors with a forward-looking perspective. However, weakness of these studies is that they are vulnerable to confounding events and require significant assumptions about the functioning of stock markets (efficiency, rationality and absence of restrictions on arbitrage).
- Accounting studies. Estimating returns from reported financial statements provides a credible indirect measure of economic value creation. Beside the possible incomparability of data for different years, companies and countries important downsides of these studies are a backward-looking perspective and ignoring the value of intangible assets.
- Surveys of managers. Insights into value creation that may not be known in the stock market are strength of studies that present a sample of executives with a standardized questionnaire. At the same time respective

methodology characterized with a necessity to recall historical results and a low rate of participation makes this research approach vulnerable to criticism of generalizability.

- Clinical studies. Case studies are not suitable to hypothesis testing due to small number of observations. However, an inductive research with objective and in-depth information often induces new insights and unveils new patterns and behaviours.

This research is focused on the influence of takeovers on profitability of target companies and uses accounting studies approach that examines reported financial results, therefore it belongs to a niche of studies on M&A which is not as extensive as other research niches on the same subject. Since performance of target companies have been compared to performance of peer companies it is important to mention several studies with similar methodology such as the analysis of a long-term profitability of corporate takeovers of both the acquiring and target companies performed by Martynova, Oosting and Renneboog (2006). Further, Ghosh (2001) compared the operating performance of merging firms to industry-median firms while Akben-Selçuk (2008) searched for changes in the corporate performance of public companies which were targets of cross-border acquisitions.

Regarding analysis performed on Croatian companies there are several studies on M&A observed on an aggregate level but researches on a company level are relatively rare. Šunje, Rahimić and Filipović (2012) focused on the importance of the compatibility in management style in acquired companies. However, along with a research done by means of a questionnaire they analyzed financial results of 43 Croatian target companies acquired in the period from 1998 until 2006. An insight into comparison of selected financial indicators (EBITDA/SALES, EBIT/SALES, ROS, ROA and ROE) in the year of a takeover with the average of their value in three years following the takeover indicated that almost 70% of analysed companies performed better after the takeover. Vretenar (2012) explored post-M&A operating performance by measuring the change in assets, equity capital, revenues, number of employees, labour productivity and profit/losses for a sample consisting of 40 Croatian publicly listed companies taken over in the 2001-2007 period. Data was extracted for a seven years period (or shorter if there was no data available) in a way to cover for the two years before the M&A and five years following the M&A. Regarding profit, the analysis of the change in profit margin showed profitability enhancement for the majority of the firms with big losses and a profitability decrease for the entire sub-sample group of firms with the highest earnings prior to takeover. Although both previous studies deal with profitability of target companies, to the authors' knowledge only Višić (2013) performed a research on Croatian companies which is methodologically similar to the one presented in this paper. The effect of takeover on company's profitability was analyzed by using several profitability measures and the research has been performed on the sample of 70 Croatian companies which were taken over in the period from year 2003 to 2008. The results emphasize the sensitivity of results on manipulation in terms of used methodology and indicators. This research has proven that the same sample and the same reference period can lead to significantly different conclusions.

Along with three previously mentioned studies, several other studies have analysed the impact of M&A activity on Croatian companies as well, but their main focus and methodology is different than the one presented in this paper. In that manner, Gonan Božac and Šumberac (2009) used a case study approach to analyse the impact of a merger on three involved companies. Although they provided an insight into a change of data in profit and loss statements, their main goal was to show the possibility of transition to a single board system when merging joint stock companies. Filipović (2012) explored the impact of company's size on takeover success. However, he used a questionnaire to test the opinion of 43 board members and company managers regarding post-takeover performance of target companies. Further, by applying logistic regression on a sample of 50 acquired listed and privately held companies, Pervan et al. (2010) revealed that the likelihood of being acquired is related to the variables such as EBITDA margin, activity, size, liquidity and leverage. Also, Filipović, Cingula and Tipurić (2011) tested whether companies with a below average profitability in its industry are more likely to become takeover targets than other companies in the industry while Pavić Kramarić (2012) explored characteristics of Croatian target insurance companies.

3. Influence of M&A activity on companies' performance

In order to investigate the influence of mergers and acquisitions on firms' performance a sample of 116 companies that were acquired between 2008 and 2011 was created. Identification of acquired companies (belonging to different industries) was accomplished by a Zephyr database. The collected data for each company were referred to a three-year period i.e. to the year when the takeover occurred, one year before and one year after the takeover. The sources of the collected data were financial reports (balance sheet and profit & loss statement) which were taken from the website of Croatian Financial Agency (FINA).

Since in this article we wanted to analyze the influence of M&A activity on companies' performance from different aspects, the empirical part of the research encompassed four different segments. The first part aimed to test whether the costs of the acquired companies were lower after companies engaged in M&A. The second and the third part were focused on the companies' profitability and comparison of their post-merger performance with those of pre-merger. The final segment aimed to reveal which investors (domestic or foreign) carried out most of the takeovers.

It is assumed that companies can achieve and exploit economies of scale and consequently reduce costs by merging with another company. Also, companies can achieve tax savings, reduce the cost of employees and many others costs that could not be avoided by doing separate business. In order to test the previous statement, a Paired-Samples T Test is conducted and the results of the test are presented in the Table 1.

Table 1. Results of Paired-Sample test – total expenditures

		Paired Differences							
					95% Confidence Interval of the Difference				
		Mean	Std. Deviation	Std. Error Mean	Lower	Upper	t	df	Sig. (2-tailed)
Pair 1	TEX1 - TEX3	-2,244E7	5,010E8	4,713E7	-1,158E8	7,094E7	-,476	112	,635

Source: Compiled by authors

Based on the results from the Table 1, it can be concluded that pre-merger company's total expenditures did not statistically differ from the company's total expenditures after the takeover occurred. Results remained robust even when total expenditures were replaced by operating costs.

Some empirical studies demonstrated that M&A positively affect firms' profitability (e.g. Powell & Stark, 2005: Ragman & Limmack, 2004). Such a positive impact of M&As is usually interpreted in terms of benefits resulting from larger market shares, differentiation of products and services, better management etc. In order to approve or disapprove positive influence of M&A activities on companies' business results, different measures of firms' profitability might be applied. However, when selecting appropriate indicator of corporate performance one should keep in mind different concerns of individuals interested in the companies' success, such as: owner-shareholders, top management, and other interested parties. Each of these groups has specific interests in terms of business performance, and hence we employed three different measures of profitability: return on assets (ROA), return on equity (ROE) and profit margin (PM). The results of the Paired-Samples T Test for each of the selected variables are presented in the Table 2.

Table 2. Results of Paired-Sample test – ROA, ROE, PM

		Paired Differences							
					95% Confidence Interval of the Difference				
		Mean	Std. Deviation	Std. Error Mean	Lower	Upper	t	df	Sig. (2-tailed)
Pair 1	ROA1 – ROA3	3,8471707	30,102273	2,7949260	-1,6890398	9,3833811	1,376	115	,171
Pair 1	ROE1 – ROE3	206,90773	1,93260E3	179,4380261	-148,5244649	562,3399256	1,153	115	,251
Pair 1	PM1 – PM3	1,9488E3	2,04441E4	1,8981901E3	-1,8110726E3	5,7088259E3	1,027	115	,307

Source: Compiled by authors

It is obvious from the Table 2 that there are no statistically significant differences between pre-merger and post-merger companies' profitability. These results motivated us to perform additional analysis and to check whether there exists a difference between profitability of companies from our samples and those of peer companies (i.e. similar companies that operate in the same three-digit industry level; that are similar in size and have similar operating results (up to +100% and -50%), but have not experienced M&As). For this purpose, some companies for which peer company couldn't be found had to be omitted from further analysis. The number of the remaining companies in the sample was 71, thus suggesting an inclusion of 71 peer companies for which data was collected from Amadeus database. Even when profitability measures of sample companies were confronted with those of peer companies, results regarding statistically significant differences between them remain the same – there was no significant difference (Table 3). These results are in accordance with those obtained by Sharma & Ho (2002) and Ghosh (2001).

Table 3. Results of Paired-Sample test – peer company

		Paired Differences							
					95% Confidence Interval of the Difference				
		Mean	Std. Deviation	Std. Error Mean	Lower	Upper	t	df	Sig. (2-tailed)
Pair 1	ROA3 - xROA3	-50,01120	431,83234	52,3673636	-154,5368973	54,5144934	-,955	67	,343
Pair 2	ROE3 - xROE3	55,847596	2,67599E3	326,9246361	-596,8784997	708,5736925	,171	66	,865
Pair 3	PM3 - xPM3	34,292464	635,96651	77,6956762	-120,8319583	189,4168870	,441	66	,660

Source: Compiled by authors

Final part of the empirical research was focused on the structure of the acquirers according to their residence country. For the Republic of Croatia, as well as for other small countries, an economic rule according to which the basic prerequisite for the economic development is an achievement of higher degree of openness of the economy, or in other words the internationalization of business, is undoubtedly valid. For Croatia, this provided an entrance of foreign investors who mainly through the acquisitions of domestic enterprises have contributed to their salvage. Figure 1 gives an insight into the structure of the acquirers according to their residence country.

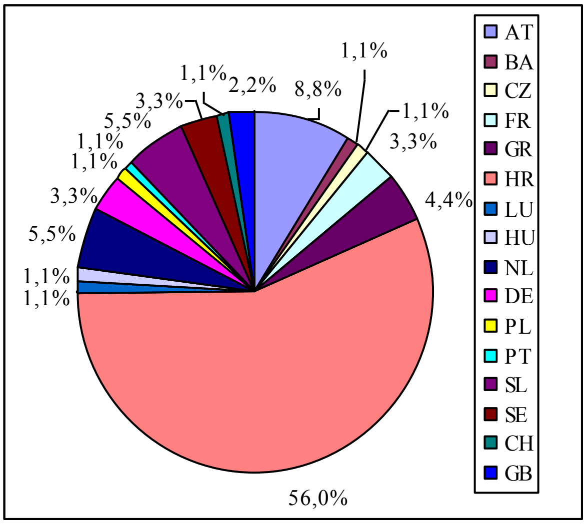

Fig. 1. Structure of acquirers by country

Although one might expect that the majority of acquisitions would be conducted by foreign investors, the Figure 1 reveals that in 56% of cases the residence country of acquirers was Croatia, thus suggesting a larger number of takeovers carried out by domestic than by foreign companies. This may be because domestic firms, compared to foreign ones, are still more familiar with the market situation, Croatian legislation, domestic competition, regional business trends, market needs, the political environment, etc.

4. Conclusions

The aim of this research was to analyze the influence of M&A activity on companies' performance. In order to accomplish this objective, a sample of 116 companies that were acquired between 2008 and 2011 was created. Acquired companies were identified by a Zephyr database. Results of the conducted analysis showed that: (1) the costs of the acquired companies weren't lower after companies engaged in M&A; (2) there are no statistically significant differences between pre-merger and post-merger companies' profitability; (3) even when the profitability measures of sample companies were confronted with those of peer companies, results remained the same, i.e. there were no statistically significant differences between them; (4) majority of acquisitions were carried out by domestic companies. Further research on the respective subject will include cross-country and/or cross-industry analysis.

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